Introduction to Software Project Management

Software

Project

Software Project

Engineering

Management

Software Engineering

Software Project Management

Introduction to Software Project Management contd..

- Software (IEEE): collection of programs, procedures, rules, and associated documentation and data

- Project: A Project is a temporary endeavor or effort undertaken to create a unique product, service or result

Engineering

- The American Engineer’s Council for Professional Development has defined "engineering" as

  The creative application of scientific principles to design or develop structures, machines, apparatus, or manufacturing processes,

  or works utilizing them singly or in combination; or

  to construct or operate the same with full cognizance of their design; or to forecast their behaviour under specific operating conditions; all as respects an intended function, economics of operation or safety to life and property.

Management

- **Management** is the function that coordinates the efforts of people to accomplish goals and objectives using available resources efficiently and effectively.

- Management word is generally used with respect to business and organizations

Management contd..

- Planning - decides what is to be done
- Organizing - making arrangements
- Staffing - selecting the right people for the job
- Directing - giving instructions
- Monitoring - checking the progress
- Controlling - taking action to remedy hold ups
- Innovating - coming up with solutions when problems arise
- Representing - liasoning (act as a link) with clients, users, developers and other stakeholders

Software Engineering

- s/w Engg. focuses on general software
- s/w Engg. (IEEE): systematic approach to development, of software operation, maintenance, and retirement of sw
- Systematic approach: includes methodologies and practices that can be used to solve a problem
- The problem of producing software to satisfy user needs drives the approaches used in SE

Project Characteristics

- Temporary

Projects starts and ends

Project is completed when the objectives are achieved

Examples

- Monument creation
- Artefacts Production
- Research Project producing a document is useful for decision making

Project Characteristics contd..

- Duration

  (generally 6 months to 10 years)

  2 or 3 years are very common time periods
- Uniqueness (Every project has different objectives)
- Scope (can not be very general)

Project Characteristics contd..

- Can not be well defined in the beginning (progressive elaboration is possible)

  Work carried out for customers
- Large and complex projects involve many disciplines, and number of phases
- Time and Resource constraints

Difference between software projects and other projects

- Invisibility (You may not see the result as a product)
- Complexity
- Conformity or consistency (scientific methods are followed)
- Flexibility (can be modified easily)

Projects versus jobs

- A job is a sequence of specific operations
- A job includes a number of routine tasks
- Each task is well defined and well understood
- Uncertainty is negligible

Software for Industry

- Others who have not developed the software are the users
- Requirements
  - bugs not tolerated
  - Good user Interface is required
  - Documents needed for the user as well as for the organization and the project

Software for Industry contd..

- Supports important functions for business
- Reliability , robustness are very important
- Heavy investment
- Portability is a key issue

Software for Industry contd..

- Brooks thumb-rule: Industrial strength sw costs 10 times more than other sw
- Domain of SW Engg: Industrial strength sw
- In SW Engg. and in this course, software means industrial strength software

Driving factors for software

Productivity (the effectiveness of productive effort)

- An engineering project is driven by cost and schedule
- In s/w, cost is mainly manpower cost, hence measured in person-months
- Schedule is in months/weeks – very important in business context
- Productivity P captures both of these
  - If P is higher, cost is lower
  - If P is higher, time taken can be lesser

Driving factors for software contd..

Quality (Q)

- Developing high Q s/w is a basic goal
- Quality of s/w is harder to define
- Approaches used should produce a high Q software

Quality: ISO standard

- Requires six attributes
  - Functionality
  - Reliability
  - Usability
  - Efficiency
  - Maintainability
  - Portability

Reliability

- Reliability is generally considered the main Quality criterion
- Reliability is a measure of of failure
- = 1 - Probability of failure
  - hard to measure
  - approximated by no. of defects in software

Basic approach of SE is to separate process from products and focus on process and managing the process

Software Project

- Goal: Build a software system to meet commitments on cost, schedule, quality
- Worldwide - many projects fail
  - one-third are due to cost or schedule overrun

Project Management

- **Project management** is the process and activity of planning, organizing, motivating, and controlling resources, procedures and protocols to achieve specific goals in scientific or daily problems.

Objectives of a project

- **Project Manager:** A project is taken care by a project in-charge or manager (may be a person or a group) who sets the project scope and allocates costs
- The group may be known as a project board, management board or the steering committee
- Focus of project manager or project board is on what will be done rather than how it will be done

Objectives should be SMART

S Specific, concrete and well defined

M Measurable, Can be objectively judged

A Achievable (within the power of the individual or the group to meet the target)

R Relevant to the purpose of the project

T Time constrained (within sometime the objectives should be achieved)

Goals or sub-objectives

- Steps to achieve objectives are known as goals
- Objective X will be achieved if the sub-objectives or goals A, B, C etc. are all achieved

- Goal can be allocated to an individual but not the objective

Example of objective and goal

Objective: User satisfaction with software product

Analyst goal: Accurate requirements

Developer goal: software that is reliable

Measures of effectiveness: How do we know goals or objectives have been achieved?

By practical test they can be objectively assessed

For example, if the number of complaints are low and the customers buy more products, then user satisfaction is achieved

Formal Project Management

- Formal Project Management and Formal Development methods play an important role in the execution of large and complex projects than small projects
- Formality: well defined processes used for each task; measurements used to control

Product definition

- A **product** is the item offered for sale. A **product** can be a service or an item. It can be physical or in virtual or cyber form.

(economictimes.indiatimes.com)

- A good, idea, method , object or service created as a result of a process and serves a need or satisfies a want.
- It has a combination of tangible and intangible attributes (benefits, features, functions and uses that a seller offers a buyer for purchase).
- For example a seller of a toothbrush not only offers the physical product but also the idea that the consumer will be improving the health of their teeth.

(http://www.businessdictionary.com)

Product Contd..

- A project when successfully completed and made operational or converted into something useful is a product
- Categories of projects

  - Information systems versus embedded systems

  - Objective based versus product based

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